

What determines the level of the asset swap spread? It is a function of the relative pricing of the agency security versus the swap curve. It is a function of both the *credit spread* of the bond (defined as the difference between the yield of the bond and the yield of the relevant Treasury benchmark) and swap spreads. Two important facts to keep in mind are as follows:

- As the credit spread of the asset increases (i.e., the asset becomes cheaper), the asset swap spread increases as well.
- As swap spreads increase, the asset swap spread decreases.

To gain intuition for these two facts, refer again to Figure 8.20. If the credit spread of the asset increases (everything else the same), then this will decrease the dirty price of the bond. A decrease in the dirty price decreases the upfront fee that the swap dealer has to make. As the upfront fee decreases, the swap dealer can afford to pay a larger spread to LIBOR, and hence the asset swap spread increases. If, on the other hand, swap spreads increase (everything else the same), then swap rates increase, and the 6% fixed coupon that the investor is paying to the dealer becomes “worth less.” That is, the value of receiving a 6% fixed coupon declines as swap rates increase. If the dealer is receiving fixed flows that are worth less, it will pay out a lower spread to LIBOR to compensate. So as swap spreads increase, asset swap spreads decrease. Thus, the asset swap spread depends on the relative pricing of the credit spread of the underlying security and the swap spread that matches the maturity of the bond.⁵¹

8.4.1 Asset Swap Math

The asset swap spread is the spread to LIBOR that makes the swap outlined in Figure 8.20 an NPV = 0 transaction. In this section, we will give some further insight into how exactly that spread is determined. Let us assume, without any loss of generality, that the investor and the swap dealer exchange principal at maturity in the terms of the swap. As we know, this will not change the value of the swap at all.

Let’s take a look at the swap now. The investor receives floating payments (LIBOR plus a spread) plus an upfront payment equal to the dirty price minus par. The present value of these flows can be written as

$$PV(\text{Floater}) + PV(\text{Spread}) + \text{Dirty Price} - \text{Par}, \quad (8.4)$$

⁵¹Recall that in Section 8.3 we said that some would regard a seemingly macro swap spread trade as a relative value trade. Hopefully this discussion sheds some light on why that interpretation is reasonable. If we put on a swap spread widener, then we are buying Treasuries and paying fixed. Recall that an asset swap is a trade in which we buy a security and pay fixed, and thus a swap spread trade is really an asset swap.

where Floater represents all the floating coupons and principal at maturity. The investor pays fixed coupons and principal at maturity, the present value of which we can write as

$$\text{PV}(\text{Fixed Bond}), \quad (8.5)$$

where Fixed Bond represents all of the fixed coupon payments in the swap plus principal at maturity. Since the swap is an $\text{NPV} = 0$ transaction, the present value of the cash flows coming in must equal the present value of the cash flows going out. Equating Equations 8.4 and 8.5 gives

$$\text{PV}(\text{Floater}) + \text{PV}(\text{Spread}) + \text{Dirty Price} - \text{Par} = \text{PV}(\text{Fixed Bond}). \quad (8.6)$$

Recognizing that $\text{PV}(\text{Floater}) = \text{Par}$, we can simplify Equation 8.6 and write

$$\text{PV}(\text{Spread}) = \text{PV}(\text{Fixed Bond}) - \text{Dirty Price}. \quad (8.7)$$

The dirty price of the bond is of course that price at which the bond is trading in the market. It is a function of discounting the cash flows of the bond implicitly using its own credit curve. The $\text{PV}(\text{Fixed Bond})$ represents taking those same exact cash flows but instead discounting them using discount factors generated by the swap curve. Some people refer to the $\text{PV}(\text{Fixed Bond})$ as the “LIBOR price of the bond.”

To say all of this another way, the dirty price of the bond is the actual price of the bond; it is the price of the bond as determined by the market. The $\text{PV}(\text{Fixed Bond})$, or LIBOR price of the bond, answers the following question: if we take all the cash flow from the bond (including principal) and discount those cash flows in accordance with zero coupon rates from the swap curve, what price do we get? It is the price of the bond according to the swaps market. As shown in Equation 8.7, the “purpose” of the spread is to reconcile these two different prices. That is, the bond market says the bond is worth one amount; the swaps market says if we take those same cash flows from the bond and discount them using the swap curve, we get a different amount; and the spread makes up the difference between those two amounts.

Let’s go back now to the asset swap shown in Figure 8.20. Again, this was a 4-year agency security paying coupons of 6%, and we said that it asset swaps to $\text{LIBOR} - 11.9$. In the top half of Figure 8.21 we showed the calculation of the asset swap spread. In the bottom half of figure 8.21 we show that the difference between the LIBOR price of the bond and the dirty price of the bond is $\$101,866,152 - \$102,286,252 = -\$420,100$, or -42 bps of the notional. And we also show that -42 bps is indeed the present value of the asset swap spread of -11.9 bps running.

8.4.2 Asset Swaps Today

We just spent a considerable amount of time going over par asset swaps, but, as mentioned, this methodology of executing asset swaps is no longer in favor. There are several reasons why this style of asset swap has gone out of vogue. The main reason is the upfront fee associated with a par asset swap. Let's say an investor does a par asset swap of a high coupon, long maturity Treasury. The dollar price on this bond could very well be around \$150. So if the investor does a par asset swap on, say, \$100mm of bonds, that means that the swap will have an upfront fee of \$50mm. It also means that once this fee is paid, the remaining swap will be around \$50mm off-market. This creates significant credit exposure for the party who has paid the upfront fee. Although many counterparties have a Credit Support Annex in place with one another that specifies the transfer of collateral against the mark-to-market of swaps so as to mitigate credit risk, the transfer of collateral can be costly and time-consuming.

It is nonetheless worthwhile to go through the mechanics of par asset swaps if for no other reason for its historical context. Moreover, the analysis of the par asset swap lends itself well to understanding the inherent position being put on in any asset swap, namely, the relative movement of the credit spread of the bond versus swap spreads.

Today's favored asset swap transaction is actually much simpler. Referred to as a *yield-yield asset swap* (and sometimes as a *matched-maturity asset swap*), it is accomplished by buying a bond and paying fixed on a spot starting swap whose maturity matches the maturity of the bond.⁵² There is no upfront fee in the swap, and there is no spread to the LIBOR side. The coupon on the swap is the par swap rate for that maturity; that is, the coupon is the rate that gives the swap an NPV = 0. This method of asset swap is typically much faster to execute in the market.

Example

Consider our stylized example and suppose that the swap curve is as given in Table 8.1. Consider an investor who buys the 4-year agency bond at a price

⁵²Thus, it is often the case that the swap in a yield-yield asset swap does not have a maturity date that falls an exact number of years (or an exact number of semiannual periods) after the effective date. In this case, the swap will typically be structured with a *short stub* upfront on both sides, meaning the first fixed flow will occur in less than six months and the first floating flow will occur in less than three months. The first fixed coupon will occur at the same time as the next coupon payment of the bond. The first floating flow will occur at this time as well, unless that happens to be more than three months from the effective date. In this case, the first floating flow will occur three months prior to the first fixed payment.

of \$102.286252 that corresponds to a yield of 5.35%. To execute a yield-yield asset swap of this bond, the investor would pay fixed in four years at the par swap rate, which is 5.47%. The yield-yield asset swap spread is simply the difference between the bond yield and the swap rate, or $5.35\% - 5.47\% = -12$ bps.⁵³ $\square$

More generally, in a yield-yield asset swap, the asset swap spread is defined to be

$$\text{Yield-Yield Asset Swap Spread} = \text{Bond Yield} - \text{Swap Rate}, \quad (8.8)$$

where Bond Yield is the yield of the bond, and Swap Rate is the par swap rate on a spot starting swap that has the same maturity as the bond.⁵⁴ In the event that the Bond Yield and the Swap Rate are quoted as spreads to the same Treasury benchmark, then Equation 8.8 reduces to

$$\text{Yield-Yield Asset Swap Spread} = \text{Credit Spread} - \text{Swap Spread}, \quad (8.9)$$

where Credit Spread is the spread, in bps, that is added to the yield of the relevant Treasury benchmark to come up with the Bond Yield, and Swap Spread is the number of bps that is added to the yield of the relevant Treasury benchmark to come up with the swap rate.⁵⁵ From Equation 8.9, we see, then, that a yield-yield asset swap is also a measure of the relative value of the credit spread of the bond to the swap spread of the same maturity. As the Credit Spread increases (i.e., the bond cheapens), the Yield-Yield Asset Swap Spread increases (i.e., the asset swap cheapens), and as the Swap Spread increases, the Yield-Yield Asset Swap Spread decreases (i.e., the asset swap richens). These are the same dynamics we observed with the par asset swap transaction.

⁵³Notice that the level of the yield-yield asset swap spread, -12 bps, is very close to the par asset swap level of -11.9 bps that we found earlier. Oftentimes, these levels are very close to one another. In fact, a good handcheck (in many situations) when calculating the par asset swap spread is the level obtained from the yield-yield asset swap.

⁵⁴Most often both the Bond Yield and the Swap Rate are quoted on a semiannual 30/360 basis.

⁵⁵In the case where both the bond and the swap are quoted over the same Treasury benchmark,

$$\begin{aligned} &\text{Bond Yield} - \text{Swap Rate} \\ &= (\text{Treasury Yield} + \text{Credit Spread}) - (\text{Treasury Yield} + \text{Swap Spread}) \\ &= \text{Credit Spread} - \text{Swap Spread}. \end{aligned}$$

In the case in which the bond and the swap are quoted over different Treasury benchmarks, the Treasury yields do not cancel out. In this case, the yield-yield asset swap spread is a function not only of the difference between the Credit Spread and the Swap Spread but of a (typically de minimis) Treasury curve component.

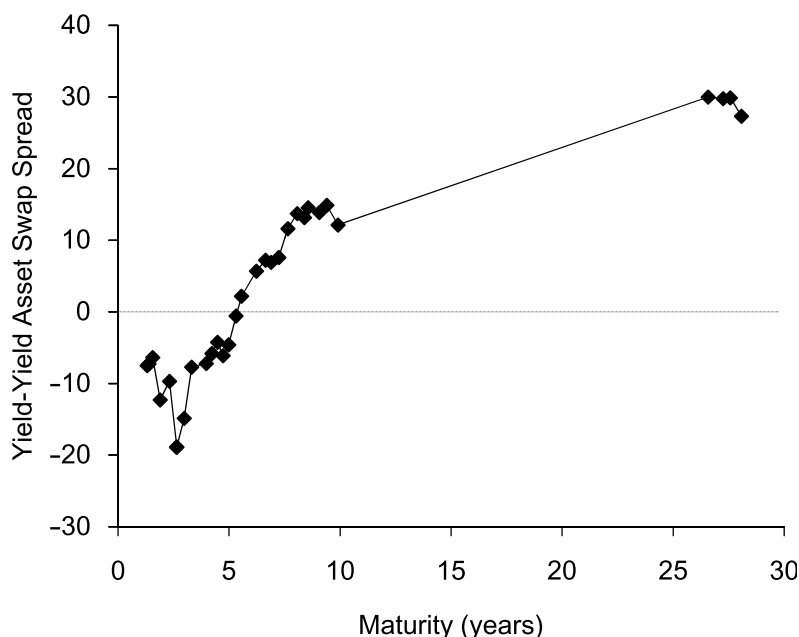


Figure 8.22: The Asset Swap Curve for FNMA Debt, October 25, 2002

FNMA Asset Swaps

As mentioned previously, one very popular type of bond to asset swap is agency fixed rate bullet debentures.⁵⁶ If an investor buys an agency bond on asset swap, he is not taking an outright position in the credit of the underlying agency but, rather, he is taking a position on agency spreads relative to swap spreads. Pretty much the entire universe of agency non-callable debt is looked at routinely on an asset swap basis by a myriad of investors and dealers. Figure 8.22 shows the *yield-yield asset swap spread curve* for a sampling of Federal National Mortgage Association (FNMA) bullet debentures using data from October 25, 2002. That is, this graph shows the relationship between the asset swap spread of FNMA debt (as defined in Equation 8.8) and maturity. In 2002, FNMA actively issued securities with a variety of maturities of ten years and less but was no longer issuing benchmark debentures with a maturity of greater than ten years (and the last time it had done so as of this time was when it had issued a 30-year bond, the 6.625%

⁵⁶Some have suggested that liquidity in certain agency debentures, both on an outright basis as well as on asset swap, has diminished since the financial crisis.

of November 15, 2030, in 2000.) As of this date, FNMA had no benchmark issues with a maturity between ten and 26 years.

As this figure demonstrates, the FNMA asset swap spread curve has traditionally been upward sloping (albeit with a kink near the front end of the curve), indicating that longer-dated maturity securities are cheaper on a LIBOR basis than are shorter-dated securities. At the time at which this data was compiled, FNMA securities traded “LIBOR minus” (meaning their asset swap spreads are less than zero) for securities that had a maturity of about five years or less and traded “LIBOR plus” for longer maturities. Investors take views on the relative richness or cheapness of particular bonds on asset swap and trade them accordingly.⁵⁷

FNMA Versus FHLMC Asset Swaps

Historically, the debt issued by FNMA has asset swapped at nearly the same levels as the debt issued by the Federal Home Loan Mortgage Corporation (FHLMC). Figure 8.23 provides a comparison of FNMA asset swap spreads on its benchmark on-the-run 10-year debt versus asset swap spreads on FHLMC 10-year on-the-run benchmark debt during 2002 when concerns mounted about FNMA’s risk management policies.⁵⁸ Traditionally, FNMA had been considered by many in the market a slightly stronger credit than FHLMC, and this manifested itself in terms of the slightly richer asset swap levels for FNMA often observed (i.e., the difference between FNMA’s asset swap spread and FHLMC’s asset swap spread was negative). However, in 2002 FNMA reported extreme levels for its *duration gap*, calling into question its risk management processes in the minds of some investors.⁵⁹ During this period of time, FNMA debt cheapened to FHLMC on an asset swap basis. As shown in this figure, the difference between FNMA’s asset swap

⁵⁷ An investor who believes that a bond is cheap on asset swap (i.e., believes that the asset swap spread will decline) will want to buy the bond and pay fixed on the swap. An investor who owns a bond on asset swap and believes the asset swap is rich (i.e., believes that the asset swap spread will go up) will want to sell the bond and unwind the swap.

⁵⁸ During this time, both FNMA and FHLMC routinely issued bullet debentures of a variety of maturities on a regular schedule. For a given maturity series, the most recently issued benchmark bond maintained an “on-the-run” status, much as on-the-run Treasuries do.

⁵⁹ FNMA’s duration gap, the difference between the duration of its asset and the duration of its liabilities, was viewed by many market participants as a measure of its interest rate sensitivity. The duration gap was measured in months, and in 2001 FNMA’s stated objective was to keep the gap between –6 months and +6 months. In three consecutive months starting in July 2002, the duration gap was –9 months, –14 months, and –10 months, raising concern among many investors. See Frame and White (2004) for more details.

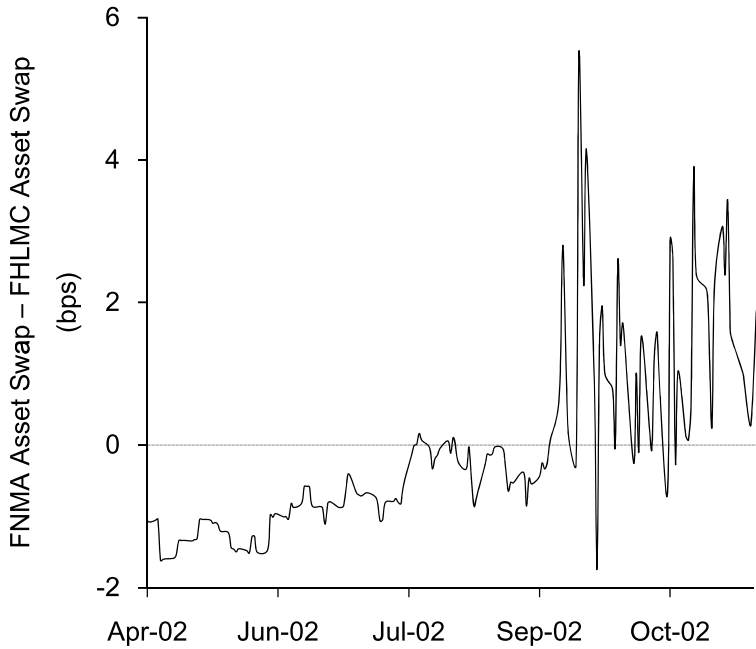


Figure 8.23: FNMA Versus FHLMC Rolling 10-Year Benchmark Asset Swap Spreads

spread and FHLMC's asset swap spread turned positive, indicative of the relative cheapening of FNMA to FHLMC.

FNMA Asset Swap Spreads Before and After Lehman Fell

Let's refer back now to the asset swap spread curve for FNMA, such as the one depicted in Figure 8.22. Although there is definite variation in the asset swap spreads of particular bonds from one day to the next, the general shape and level of the asset swap spread curve for FNMA remained more or less stable for many years. For example, Figure 8.24 shows the FNMA asset swap spread curve for data taken in March 2008. Both Figure 8.22 and 8.24 look rather similar, with the most expensive bonds asset swapping around -20 , and the cheapest bonds asset swapping around 30 to 35 bps.⁶⁰

⁶⁰This is not to suggest that asset swap levels on agency bonds never change. On the contrary, it was not uncommon during this time for asset swap levels to have daily fluctuations of several bps. Nevertheless, the general shape and levels depicted in these two figures were commonplace. Market participants take views on the levels of asset swap

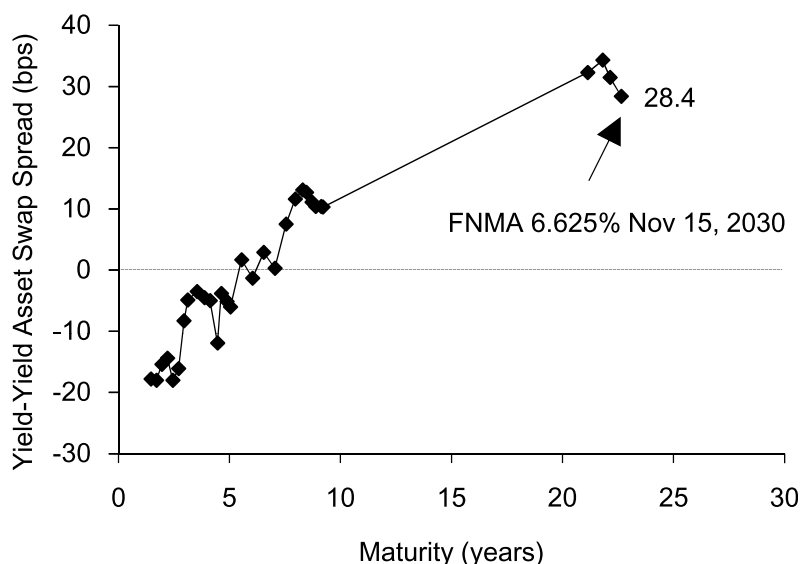


Figure 8.24: The Asset Swap Curve for FNMA Debt, March 31, 2008, with Data Label Indicating Asset Swap Spread of FNMA 6.625% of November 15, 2030

In Figure 8.24 we have placed a data label next to FNMA's longest-dated bullet debenture at this time, the 6.625% of November 15, 2030. This bond, which had approximately 22.5 years until maturity at the time, had a yield-yield asset swap level of 28.4 bps on this date.

The agency asset swap curve changed dramatically after Lehman fell in the fall of 2008. Figure 8.25 shows the asset swap curve for FNMA using data from April 2009. This figure shows that the asset swap curve during this time had the same basic shape as it did in the other two periods we discussed. The startling thing to notice, however, is that the asset swap spread for long-dated securities was nearly 100 bps higher than it had been historically. There is a data label next to the 6.625% of November 15, 2030 which, at the time, was a bond with approximately 21.5 years until maturity. It indicates an asset swap level of 112 bps for this security, as compared to 28.4 bps in March of the previous year. This was an unprecedented move in asset swap spreads. It's important to remember that an asset swap spread

spreads, especially when they diverged from their long-term means, usually looking to make at least a few bps on the trade. For trades held for longer holding periods, the implications of carry and roll-down are also a consideration, just as with many of the other trades we have looked at in this chapter.

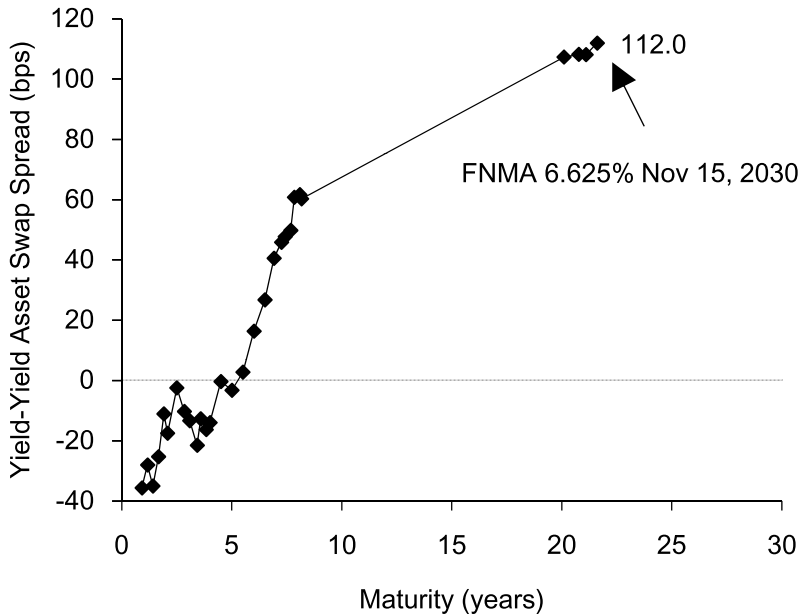


Figure 8.25: The Asset Swap Curve for FNMA Debt, April 13, 2009, with Data Label Indicating Asset Swap Spread of FNMA 6.625% of November 15, 2030

depends on the difference of two different spreads (i.e., the credit spread of the underlying bond versus the swap spread of the relevant maturity swap). The dramatic move in the asset swap spread of this longer-dated paper was a result of two phenomena that were going on during this period of time:

1. During the crisis, credit spreads in general widened out significantly. In particular, long-dated *agency spreads* (the difference between the yield on these securities and the Treasury benchmark to which they are compared) widened out significantly.
2. At the same time, long-end swap spreads tightened dramatically. For example, 30-year swap spreads tightened from 34.5 bps on March 31, 2008, to -27.75 bps on April 13, 2009.⁶¹

Keeping Equation 8.8 in mind, Table 8.10 provides an explanation as to why

⁶¹As mentioned in Chapter 1 and as will be discussed further in Chapter 9, 30-year swap spreads traded in negative territory following the collapse of Lehman, the first time in history USD swaps spreads ever traded below zero.

Date	Agency Yields	Matched Maturity Swap Rates	Yield-Yield Asset Swap Levels
March 31, 2008	$T_{30} + 58$	$T_{30} + 29.6$	$T_{30} + 58 - (T_{30} + 29.6) = 28.4$
April 13, 2009	$T_{30} + 83$	$T_{30} - 29$	$T_{30} + 83 - (T_{30} - 29) = 112$

Table 8.10: Comparison of Yield-Yield Asset Swap Spreads of FNMA 6.625% of November 15, 2030, Before and After Lehman Fell

these two phenomena (i.e., the widening of agency spreads and the narrowing of swap spreads) contributed to a move of nearly 100 bps in the asset swap spreads of long-dated agencies, by looking at the FNMA 6.625% of November 15, 2030, in particular.⁶²

⁶²The yield of the FNMA 6.625% of November 15, 2030, was, during this period of time, quoted as a spread to the then-current on-the-run 30-year Treasury yield. On March 31, 2008, these agency bonds yielded 4.892%, and the then-current 30-year Treasury yielded 4.312%, implying a spread to Treasuries of $4.892\% - 4.312\% = 58$ bps. But on April 13, 2009, the agency bonds were yielding 4.515%, and the then-current 30-year Treasury yielded 3.685%, implying a spread of 83 bps. Thus, we saw a substantial widening of agency spreads over this period.

The matched maturity swap for this bond is a spot starting swap that matures on November 15, 2030. On March 31, 2008, this means the swap was approximately 22.5 years long. The par swap rate for this swap was 4.608%. Although not conventional (we would normally think about spreading this rate to an interpolated Treasury yield), we can think of this rate as a spread to the then-current 30-year Treasury yield of 4.312%, or $4.608\% - 4.312\% = 29.6$ bps. On April 13, 2009, the matched maturity swap was about 21.5 years long. The par swap rate for this swap was 3.395%, which we can also think about spreading to the then-current 30-year Treasury yield of 3.685%, or $3.395\% - 3.685\% = -29$ bps. Thus, whereas there was a substantial widening of agency spreads over the period, there was an even greater decline in swap spreads. These two effects resulted in the large move in the yield-yield asset swap spread.